

# Technical Report Guidance

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## 1. THE MENG PROJECTS

Projects are an important part of the degree course, they give students an opportunity to apply the knowledge learned in the rest of the course to current industrial and research problems. They also help to develop important project management, team working, organisation, and communication skills that are highly valued by employers and international research groups.

Projects are carried out under the direction and guidance of members of the academic staff and their research groups or industrial collaboration. They are important pieces of work in the degree programme. They provide the opportunity for you to demonstrate independence and originality, to plan and organise a large project over a long period, and to put into practice some of the techniques you have been taught throughout the course. Whatever your level of academic achievement so far, you can show your independence and inspiration in this project. It could be the most satisfying piece of work in your degree.

## 2. PROJECT ASSESSMENT

It is important to understand the way your project will be assessed. A good first-class project involves a combination of sound background research, a solid implementation or piece of theoretical work, and a well-structured and well-presented report detailing the project's background, objectives and achievements. The very best projects invariably cover some new ground, e.g. by developing a system which does not already exist or by enhancing some existing system, application or method to improve its functionality, performance etc.

A completely straightforward implementation project is unlikely to gain first-class marks, regardless of how well it is done. Similarly, projects which are predominantly survey reports will not gain high marks unless they are backed up with good **description** of experimentation, implementation, or theoretical analysis, supported by **justification, analysis and evaluation** (e.g. for performing an objective comparison of the surveyed methods, techniques etc.)

The following components will be taken into consideration:

### 2.1 Report - Presentation & Clarity (5%)

This assesses your ability to clearly and concisely communicate your work and your findings to other people. The report is the single most important component of the project and you should invest a considerable amount of time in its preparation. You should read the 'Project Report' section below to get a feel for what is required.

### 2.2 Background Preparation. (5%)

This component assesses the way you arrived at your initial project specification, work programme and list of objectives. It particularly addresses the background research undertaken and the manner in which your approach and programme of work fits in with the current state-of-the-art.

### 2.3 General Competence. (10%)

This assesses your overall approach to the project, planning and your ability to overcome the inevitable complications which arise. The specific areas in which you will be assessed are management and organisation, reliability and punctuality, overall technical competence, and your individual contribution to the project.

### 2.4 Technical achievement. (20%)

This assesses the main technical output from the project. It addresses specific issues such as the design, correctness, elegance, usability etc. of the final product and the significance of the work in relation to the state-of-the-art.

### 3. Final report

The project report is an extremely important aspect of the project and its quality will have a major influence on the final project mark. It serves to show what you have achieved and should demonstrate that:

- you understand the wider context of Electrical / Electronic / Robotic engineering;
- you can apply the theoretical and practical techniques you have been taught to the problems that you are addressing;
- you are capable of **objectively criticising** your own work, placing it in comparison with published literature, and making constructive suggestions for improvement or further work based on your experiences so far;
- as a professional engineer, you can document **clearly** your thinking and working processes for third parties who may not be experts in the field in which you are working;
- you can express this information in a **concise** manner;

With the exception of the project supervisor, the report assessors might not have followed your project throughout and for this reason will rely heavily on the report to judge the quality of your work. The same applies to the external examiners whose job it is to provide an opinion, heavily influenced by the individual project, to the exam board on borderline candidates.

Do not underestimate the importance of the report and make the mistake of thinking that top marks can be achieved simply for working hard producing a good product. This is fundamentally not the case and many projects have been graded well below their potential because of an indifferent or poor write-up. In-order to get the balance right you should consider that the aim of the project is to produce a good report, and that software, hardware, theory etc. that you developed during the project are merely a means to this end. Don't make the mistake of leaving the write-up to the last minute. Ideally you should produce the bulk of the report as you go along and use the last week or two to bring it together into a coherent document.

The physical layout and formatting of the report is also important, and yet is very often neglected. A tidy, well laid out and consistently formatted document makes for easier reading and is suggestive of a careful and professional attitude towards its preparation.

### 3.1 Report Format

|             |                                                                                            |                                                                                                                                                                                                                                                                                                                             |
|-------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Front       | Cover<br>Title<br>Group[ ]                                                                 | NO NAMES (Anonymous) Student<br>Numbers                                                                                                                                                                                                                                                                                     |
| Printing    | Paper                                                                                      | white A4                                                                                                                                                                                                                                                                                                                    |
|             | Sides                                                                                      | single sided                                                                                                                                                                                                                                                                                                                |
| Page Layout | Margins<br><i>left</i><br><i>right</i><br><i>top</i><br><i>bottom</i>                      | 40mm / 1.57"<br>15mm / 0.59"<br>15mm / 0.59"<br>15mm / 0.59"                                                                                                                                                                                                                                                                |
| Text        | Word-Processing                                                                            | Microsoft Word                                                                                                                                                                                                                                                                                                              |
|             | font<br>style<br>size                                                                      | Arial<br>regular<br>12 (normal text) : 16 (headings)                                                                                                                                                                                                                                                                        |
|             | Line Spacing<br>footnotes<br>"quotations"                                                  | <i>main text</i><br>1.5<br>1.0<br>1.0                                                                                                                                                                                                                                                                                       |
|             | Justify                                                                                    | <b>Left</b> and <b>Right</b> margins                                                                                                                                                                                                                                                                                        |
|             | Style                                                                                      | English, third person, past tense                                                                                                                                                                                                                                                                                           |
|             | Headings                                                                                   | decimal notation<br>1. Main Heading<br>1.1 Lesser Heading<br>1.1.1 Small Heading                                                                                                                                                                                                                                            |
|             | Avoid<br>abbreviations                                                                     | &, etc., e.g. , i.e., jargon, slang<br>only use abbreviations (& acronyms) if the<br>text already contains the full version<br>followed by the abbreviation in brackets.<br>e.g..Silicon Controlled Rectifier (SCR)                                                                                                         |
|             | Units                                                                                      | SI                                                                                                                                                                                                                                                                                                                          |
| Pages       | New Pages<br>Avoid                                                                         | start major sections on new pages.<br>new paragraphs and sub sections near the<br>page end                                                                                                                                                                                                                                  |
|             | Numbering<br>style<br><i>position</i>                                                      | page bottom centre<br><b>Roman</b> – up to Introduction (i, ii, iii)<br><b>Arabic</b> – from Introduction (1, 2, 3)                                                                                                                                                                                                         |
| Non Text    | Illustrations<br><br>graphs and diagrams<br><br>tables (plates)<br><br>Ensure<br><br>Avoid | number consecutively in one sequence<br>with title <b>under</b> .<br>clearly reference in text<br>LABEL: Figure #, title      TEXT: Fig.<br>#<br>LABEL: Table (Plate) #, title      TEXT:<br>Table (Plate) #<br>illustrations near to and <b>after</b> their first text<br>reference<br>placing illustrations as appendices |
| Checking    | Spellcheck<br><i>Ensure</i>                                                                | <b>English</b> selection                                                                                                                                                                                                                                                                                                    |

The report should be **concise** and not exceed **60 pages**, too much information can be as bad as not enough. This number is a limit, not a target. Reports that do not comply with this guideline are unlikely to be given a mark of more than 59% (see assessment criteria at the end of this handbook). Title page, abstract, acknowledgements, and bibliography will not count towards the word limit. Extra material can be appended to the report to allow you to disseminate all the necessary information to someone who might want to repeat your work or pick up the project details at a later stage. The appendix will not be specifically marked, but its appropriate use to disseminate all information will be judged by the assessors. See the next page for examples of the sort of content that would go in the appendix.

A typical technical or research report will have the following sections,

### 3.2 Abstract

The abstract is a very brief summary of the report's contents. It should be less than a page long (about 300 words). It should be written for a general audience, so somebody unfamiliar with your project should have a good idea of what it's about having read the abstract alone and will know whether it will be of interest to them. Include all of background, aim, method, results, and discussion/conclusion.

### 3.3 Acknowledgements

This section is not required. It is, however, usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during your project.

### 3.4 Introduction

This is one of the most important components of the report. It should begin with a clear statement of what the project is about so that the nature and scope of the project can be understood by a lay reader. After reading the introduction the reader understands what the problem is and what you intend to do about it. Is the idea to write software, or develop an algorithm, or produce hardware, or something else? Clearly state the project objectives.

You should then highlight and summarise the most interesting or important questions or problems that your project addresses, and the broader context in which those questions or problems are situated.

Finally, you must briefly introduce the structure of report (what you will cover in which chapters and how these relate to each other). You don't need to go into any detail, the aim is to make sure the reader has an idea about what will be discussed and in what order.

## 3.5 Background

You should provide enough background to the reader for them to understand what the project is all about, and what is the relevant prior work.

Examiners like to know that you have done the appropriate background research and it is important that you review either what has been done previously to tackle related problems, or perhaps what other products exist related to your deliverable. Clear references are important here, and much of this section will typically already have been written in your

You should not replicate information described in the PEP and IPL but you may reference these documents and describe any updates you have made. Use feedback from the PEP and IPL to improve what you write in your Final Report and you should note that self-plagiarism between the two reports is not possible, so no citation is needed of your own earlier writing.

- **What** does the reader need to know in order to understand the rest of the report?
- **What** problem are you solving?
- **Why** is this problem interesting or worthwhile to solve?
- **Who** cares if you solve it?
- **How** does this relate to other work in this area?
- **What** work does it build on?
- For 'research-style' projects involving the design and analysis of specific algorithms there is a large amount of relevant background both of general theory, and very specific to the algorithm you investigate.

Summarise the key findings from a range of published sources that you have used to identify research gaps, shape your aims and objectives, and justify the decisions you are making in your methodology. The text should be clear, with use of figures (with attribution) if helpful to the explanation.

Include clear aims and specific objectives and / or your hypotheses at the end of the introduction. For example, the overall aim was to do this and specific objectives were to do 1, 2, and 3. Your intro should lead to the aim: This is a problem. Something was done in the past (literature review), but not as well / enough (i.e. be critical). Therefore I am going to do something better / different aim.

## 3.6 Requirements Capture

Projects with a deliverable that serves a specific function often have an initial phase in which expected use is investigated and a brief more detailed than the specification is constructed. This would include what is necessary, what is desirable, etc in the final deliverable. The results of requirements capture determine project objectives and are used to inform project evaluation.

Requirements capture is important in all projects with real-world deliverables, and is often a significant amount of work in software projects. Where requirements capture

is less relevant (for example in an analytical 'research-style' project) this may be replaced by a detailed description of the project aims and objectives in the Introduction or the Background sections.

### 3.7 Analysis and Design

If your project involves designing a system, give a good high-level overview of your design.

In many projects, the final design is different from that originally envisaged. If the differences are interesting, write about them, and why the changes were made. Discoveries during the project may have changed the direction of work, or invalidated prior work, in which case you get credit for the design process, if it is principled, as well as the end product.

If your design was not implemented fully, describe which parts you did implement, and which you didn't. If the reason you didn't implement everything is interesting (eg it turned out to be difficult for unexpected reasons), write about it.

Note that the Project Report is written at the end of project work and must describe the project work, but need not do this chronologically. Often the best description of design, in retrospect, is far from the way in which you developed it. Where the evolution of ideas is interesting or relevant it can be described, as above, but otherwise the order in which things were done need not be documented.

The Examiners are just as interested in the engineering process you went through in performing your project work as the results you finally produced.

So make sure your report identifies when design choices have to be made, what were the possibilities, and why you made the particular choices and decisions that you did. They are looking for principled rational arguments and for critical assessment.

Engineering involves trade-offs and the reasons for a design decision may be various, and may in some cases be out of your control. Explicit understanding of this, and the ability to communicate it, is important.

Any Ethical, Legal, and Safety considerations should be described

### 3.8 Implementation

In projects with a software element give code details (not a complete listing, but descriptions of key parts). Discuss the most important or interesting aspects, and anything that was surprising or difficult. It probably won't be possible or relevant to discuss everything – state clearly what you leave out. It is important to explain the algorithm so that it is software independent (flowcharts are a good way to achieve this)



Pasting verbatim code from your project into the report is rarely a good idea; usually it should be edited down to remove extraneous details and often annotated to help the reader understand what they are looking at. Good reasons for including code could be to illustrate algorithmic flow, or highlight an interesting optimisation, demonstrate interactions with a data-structure, or give an example of input for a tool that has been designed. If you cannot explain what message or point a code fragment is conveying, it probably should not be in the report, and if you cannot justify why a line of code in the fragment helps convey that point, it maybe it shouldn't be in the code fragment.

For similar reasons, pasting screen-shots is unlikely to be a good use of space, unless it serves a particular purpose. Screen-shots are sometimes used instead of drawing a picture (for example as a “cheap” way of showing wave-forms or state machines), or in order to capture the results of running a tool (which will often be text anyway), and tend to look unprofessional and lazy. They are also sometimes used as page filler, or as “proof” that a tool was launched and something compiled, which is not necessary: your marker's default position will be to believe your claims. Use a screen-shot if it is demonstrating a particular point, such as a failure mode in a tool, or a particular interaction that is difficult/impossible to highlight, but make sure you edit and annotate the figure to show and highlight the important parts. A general rule is if there are more than one screenshot put them in the Appendix.

When discussing the implementation, it is generally best to focus on the conceptual and logical design, and only then dive into details for interesting parts or to highlight important decisions. Well thought-out figures and diagrams are often much more effective at conveying your design than source code, whether that is a data-structure, a client-server interaction, a design-flow, or a circuit hierarchy. For example, a diagram of a block-diagram of digital circuit can be used to communicate to the reader most of the important high-level decisions you have made, then VHDL fragments (or another diagram) could be used to focus on specific parts and demonstrate any interesting local details.

Complete listings may be included as appendices of your report, but this is not normally appropriate, unless the appendix is documentation, or describes an API.

Software may be provided in a separate file uploaded with your report or in a cloud repository such as Github.

No marks are awarded for the page-count of a report, and reports which contain 50 pages of printed code are more likely to be seen as showing poor judgement.

### 3.8.1 Methods

Be as detailed as possible, in that one should be able to reproduce what you did. As always, don't include unnecessary information. You should **justify every decision** you make or technique you use. You can then **evaluate** the key decisions and **analyse** their consequence on the outcome in the “Discussion”. In this section you would describe circuit and hardware, design, algorithms, simulations

## 3.9 Testing

Describe your Test Plan --how the program or system was verified. Put the actual test results in an Appendix if they are repetitive but relevant. Detailed test data may be omitted from the report if not relevant, however an accurate summary of tests should be included in the Report itself.

Sometimes non-working designs are described in project reports as though they work, when in reality they don't, or only partially work. Therefore a precise description of what works and how this has been established is important.

Examiners may try to compile, use, or test deliverables themselves (even after your report is submitted), and your report should accurately reflect the state of the project. This section is normally useful for software or hardware deliverables and less relevant in analytical projects.

## 3.10 Results

This covers an area different from the 'Testing' chapter, and relates to the understanding and analysis of the algorithm, program, or hardware behaviour. Where the deliverable is a product with easily quantified performance then the previous Chapter shows how functional correctness is established, while this Chapter shows qualitatively or quantitatively how well the product works. The list below shows typical content, not all of this will be applicable to all projects.

- An empirical relationship between parameters and results may be investigated, typically through the use of appropriate graphs.
- Where theory predicts aspects of performance the results can be compared with expectation and differences noted and (if possible) explained.
- Semi-empirical models of behaviour may be developed, justified, and tested.
- The types of experiments/simulations that were carried out should be described.
- Why were certain experiments carried out but not others?
- What were the important parameters in the simulation and how did they affect the results?
- If there are very many graphs and tables associated with this chapter they may be put in the Appendix, but it is generally better to keep these close to the text they illustrate, as this is easier for the reader.

Think carefully how you present your results so that you put the intended point across to your readers.

### 3.11 Discussion / Evaluation

This Chapter (or possibly section of the conclusions) is distinct from your results. It must contain your **critical evaluation** of your work as compared to previous analysis, algorithms, products, and when related to your original objectives. To what extent have your original objectives been fulfilled? If they have changed, what is **your rationale** for this? **What** are the advantages, disadvantages of your approach compared with related work? **How** does the scope of your work differ from related work?

Examiners expect your project report to show evidence of your ability to think as an engineer, and that includes the ability to **critically reflect** on your own work and **evaluate** its significance. For this reason it is one of the most important chapters to achieve a high score.

Material here will compare project outcomes with initial objectives and requirements captured. . Where these have changed significantly over the course of the project this should be explained and reasons given. This section should not require examiners to read your previous reports. Changes between final and initial objectives should be explained in a self-contained manner. Note that here you will reference and summarise, rather than repeat, your description of Requirements Capture earlier in the Final Report

You should also discuss the management of the project, identify areas that went well and those that did not. Identify any remedial action needed to achieve the objectives/deliverables. The reader should be able to see what you would do the same and differently if you had your time over again

You should also conduct an **objective evaluation** of the project's successes and failures and compare it to **existing literature**. It is important to understand that there is no such thing as a perfect project. Even the very best pieces of work have their limitations and you are expected to provide a proper **critical appraisal** of what you have done. Your assessors are bound to spot the limitations of your work and you are expected to be able to do the same.

Start with a quick summary of what you've done and found, i.e. 1-2 sentences, then discuss them. What do your results mean? Derive conclusions off them, but make sure they are justified. Use expressions such as 'it is likely', 'this suggests that' etc. Compare your results with literature. Discuss any limitations of the study. **Suggest improvements (what-if)** and how future work could deal with the problems you encountered. Avoid qualitative statements words such as 'very', 'good', 'little'; talk with numbers. You should back up all your achievements / findings / results with quantitative data.

### 3.12 Conclusion & Further Work

(if you want; it could be separate sections)

Give us the take-home message and how your design / findings could be used / explored further. It should clearly state how well you have met your objectives.

- How successful have you been?
- What have you achieved?
- How could the work be taken further given more time (perhaps by another student next year)?

It is important here to identify **positively** what is worthwhile in your work. At the same time, honesty, and a clear description of the limits of your work, is equally important. Take ownership of your mistakes and/or failures do not make excuses, moan or paint a negative picture

It is often most appropriate to describe work you did not have time to complete as further work.

Your readers will not be clear where, in your long report, are your most significant achievements. In the conclusions you must summarise this, referring as necessary to other sections for more detail.

- What design choices did you have along the way, and why did you make the choices you made?
- What was the most difficult and/or clever part of the project?
- Why was it difficult?
- How did you overcome the difficulties?
- Did you discover or invent anything novel?
- What did you learn?

Note that “difficult” does not necessarily mean the thing that took you the longest amount of time. Note also that the conclusions must concisely summarise this material and refer to other sections for the details.

### 3.11 References

List all sources you referenced in your report giving full details appropriately so that the reader can access each source. Information on appropriate referencing can be found in the library webpages. It is strongly advised to use a reference manager such as Mendeley; it will save you a lot of time when preparing planning and final reports as it can produce the bibliography automatically for you using the style of your liking.

### 3.12 Bibliography

Give publication details for all the citations you have made in the report. Internet URLs are allowed and, in many cases, necessary, however published material should if possible be given in standard form with publishing body, date, and author(s). Open URLs to published material, where these exist, may also be given for a reader's convenience.

### 3.13 Appendices / Annex(es)

Please use appendices as necessary for material that is tangentially relevant, or necessary to preserve project value, but that you do not expect Examiners to read.

Note that software projects with significant code should normally provide electronic versions of the code or cloud repository. In that case the Appendix should state where the code may be found.

The Appendix should normally include, or refer to, all the technical details needed for another user to continue code development. Typical items that should not be in the main report, but should (possibly -since Appendices are not normally read this is a matter of judgement) be in appendices:

Use this space for any additional information. Refer to the annex in the main text, else the reader is not going to have a reason to look at it. You could have more than one annex, as appropriate. The annexes contain information which is not essential for the 'story' to be told, but helpful to the reader that might want to dig into the detail or take your work forward. Information included here typically is program listings, user guides, complex circuit diagrams, tables, proofs, additional results, graphs or any other material which would break up the theme of the text if it appeared in the main body. Large program listings or actual files may be submitted with the report, although it is preferable either to provide them to your supervisor on a pen drive, or to cite their web path name in the report. **For group projects, an Appendix should include an indication of which group member worked on which parts of the project.**

## 4. Log Book

You will need to keep a lab book / engineer's log book. This will be submitted together to your supervisor at the end of the project. It is an important task of any engineer / researcher to document their work. Indeed, in industry, records must be kept to a very high standard as part of the GLP (good laboratory practice) or ISO 9001 accreditation of the company. In a university it is equally important as good records allow others to follow your work. For each session you go into the laboratory you should describe briefly what you did, why you did it, what you found and what that tells you. Keep a record of standard procedures, such as the standard test waveforms, designs etc

Good record keeping will save you many hours at the end of the project trying to remember what you did, what protocol you used where, and where that fantastic graph is or comes from. Written lab records should be brought to each meeting with the supervisor and submitted along with the final report.

## 5. Assessment Criteria

| Mark Range | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 90+        | <ul style="list-style-type: none"> <li>• A truly <b>Exceptional</b> project on a substantial and demanding plan of work with extremely high level of originality and initiative.</li> <li>• The project outcomes (system, theory, empirical evaluation) should be faultless, well-structured and carefully tested, proved or rigorously evaluated.</li> <li>• There should be full achievement of objectives and strong evidence of original thought exceeding expectations.</li> <li>• The project objectives must be very demanding and there should be a wide range of cogently-justified project extensions</li> <li>• The work is exemplary and is potentially publishable with minimal further editing.</li> <li>• Complex observations and evaluations of literature that are of a professional standard have been made.</li> <li>• The source material, field or laboratory work have been measured and recorded accurately, systematically and in meticulous detail.</li> <li>• Data collection and presentation conform to industry/scientific journal standards. The use of technical terminology is accurate.</li> <li>• The quality of data analysed, and literature reviewed is more than adequate to support the interpretations made and demonstrates considerable effort and outstanding use of time management throughout the project.</li> <li>• Complex interpretations have been made and have been communicated at the highest possible level. Interpretations are accurate, well-justified and show thorough knowledge of all the relevant literature.</li> <li>• Discussions and Conclusions are highly innovative, in-depth, confirm or challenge existing models and show an outstanding ability to synthesise and criticise data from a wide range of sources.</li> <li>• A thorough understanding of the work in its wider context has been demonstrated.</li> <li>• Excellent problem-solving skills and the ability to make well-reasoned independent interpretations have been demonstrated.</li> <li>• The work is concise, logically structured, grammatically correct and conforms wholly to the assessment guidelines.</li> <li>• Citations are relevant and broad in scope, and accompanying references are correct and conform to the style of an academic journal.</li> <li>• Figures are relevant, incorporate relevant and originally presented content, are of publishable quality and significantly enhance the understanding of the work.</li> </ul> |

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| 80 - 89 | <ul style="list-style-type: none"> <li>• An <b>Outstanding</b> project displaying a high level of originality, initiative, independence and thoughtfulness in conception.</li> <li>• The project outcomes (system, theory, empirical evaluation) should be essentially faultless, well-structured and carefully tested, proved or rigorously evaluated.</li> <li>• There should be full achievement of demanding objectives and some evidence of original thought above expectations.</li> <li>• The project objectives must be demanding and there should be a wide range of cogently-justified project extensions</li> <li>• The report should be well organised and presented and clearly written.</li> <li>• The work is excellent and of a publishable standard with some additional editing.</li> <li>• Very careful observations and evaluations of the literature have been made.</li> <li>• The source material, field or laboratory work have been measured and recorded accurately, systematically and with very good attention to detail.</li> <li>• Data collection and presentation conform to industry/scientific journal standards. The use of technical terminology is accurate.</li> <li>• The quality of data analysed and literature reviewed is more than adequate to support the interpretations made and demonstrate significant effort and very good time management throughout the project.</li> <li>• Complex interpretations have been made and have been communicated to a very high standard. Interpretations are accurate, justified and show good knowledge of the relevant literature.</li> <li>• Discussions and conclusions show some innovation, are in-depth, confirm or challenge existing models and show an excellent ability to synthesise and criticise data from a wide range of sources.</li> <li>• A good understanding of the work in its wider context has been demonstrated.</li> <li>• Very highly-developed problem-solving skills and the ability to make independent interpretations have been demonstrated.</li> <li>• The work is concise, logically structured, grammatically correct and conforms wholly to the assessment guidelines. Citations are relevant and accompanying references are correct and conform to the style of an academic journal.</li> <li>• Figures are relevant, mostly incorporate relevant and originally presented content, are of excellent quality and enhance the understanding of the work.</li> </ul> |
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| 70 - 79 | <ul style="list-style-type: none"> <li>• An <b>Excellent</b> project displaying originality, initiative, independence and in conception.</li> <li>• Students will show an understanding of all aspects of the project material, producing work without significant error or omission.</li> <li>• Project objectives should be reasonably demanding and fully achieved.</li> <li>• The report should display excellent organisational and presentational skills, and contain a thorough evaluation and objective critical reflection</li> <li>• The work is very good and of a publishable standard with significant additional editing.</li> <li>• Careful observations and evaluations of the literature have been made.</li> <li>• The source material, field or laboratory work have been measured and recorded accurately, systematically and with good attention to detail.</li> <li>• Data collection and presentation conform to industry/scientific journal standards. The use of technical terminology is accurate.</li> <li>• The quality of data analysed and literature reviewed is more than adequate to support the interpretations made and demonstrate good effort and good time management throughout the project.</li> <li>• Complex interpretations have been made and have been communicated to a high standard. Interpretations are accurate, justified and show sound knowledge of the relevant literature.</li> <li>• Discussions and conclusions are well-considered, in-depth, confirm or challenge existing models and demonstrate an ability to synthesise and criticise data from a wide range of sources.</li> <li>• A sound understanding of the work in its wider context has been demonstrated.</li> <li>• Highly-developed problem-solving skills and the ability to make some independent interpretations have been demonstrated.</li> <li>• The work is concise, logically structured, grammatically correct and conforms to the assessment guidelines. Citations are relevant and accompanying references are correct and conform to the style of an academic journal.</li> <li>• Figures are relevant, partly incorporate relevant and originally presented content, are of very good quality and make a valuable contribution to the understanding of the work.</li> </ul> |
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| 60 - 69 | <ul style="list-style-type: none"> <li>• A <b>Very Good</b> project should be competent in all respects.</li> <li>• The project's primary objectives are somewhat demanding and should be substantially achieved to a reasonable standard.</li> <li>• Students will show an understanding of the technical and professional issues involved.</li> <li>• The presentation and organisation of the report should be clear.</li> <li>• A well planned and executed project, written up with clarity.</li> <li>• Evidence of care and application in design, execution and reporting. <ul style="list-style-type: none"> <li>○ Alternatively, evidence of an original and demanding project but one presented without the completeness of those gaining marks at the higher bands.</li> <li>○ Within this Band, the lower mark is awarded to projects which meet many but not all of these criteria, and the higher mark is awarded to projects which meet all of these criteria, and which also meet some but not all criteria for a first.</li> </ul> </li> <li>• Good observations and evaluations of the literature have been made, but few are complex.</li> <li>• The source material, field or laboratory work have been measured and recorded accurately, but more attention to detail is required.</li> <li>• Data collection and presentation approach industry/scientific journal standards but fall short in one or more areas.</li> <li>• The use of technical terminology is mostly accurate but falls short in one or more areas.</li> <li>• The quality of data and literature reviewed is adequate to support the interpretations made and demonstrate reasonable effort and good time management throughout the project.</li> <li>• Some complex interpretations have been made and have been communicated well. Interpretations are accurate, justified and show good knowledge of the relevant literature.</li> <li>• Discussions and Conclusions show some consideration, confirm or question existing models in some aspects and demonstrate an ability to synthesise and criticise data from different sources.</li> <li>• A reasonable understanding of the work in its wider context has been demonstrated.</li> <li>• Good problem-solving skills and the ability to make some independent interpretations have been demonstrated.</li> <li>• The work is relatively concise, has a good structure, is largely grammatically correct and conforms mostly to the assessment guidelines. Some citations are not relevant and/or key citations are absent. Accompanying references are largely correct and approach the style of an academic journal.</li> <li>• Figures are relevant, partly incorporate relevant and originally presented content, are of good quality and add to the understanding of the work.</li> </ul> |
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| 50-59 | <ul style="list-style-type: none"> <li>• A <b>Satisfactory</b> project should be competent in most respects.</li> <li>• The project objectives may not be very demanding but should be achieved to a reasonable standard.</li> <li>• The presentation and organisation of the report should be reasonably clear.</li> <li>• There may be some signs of weakness, but overall the grasp of the topic should be sound.</li> <li>• A reasonably well constructed and executed project, but one which indicates a lack of attention to key features (e.g. planning, design, rationale, conducting the research, testing, data analysis, critical insight, write-up). <ul style="list-style-type: none"> <li>○ Within this Band, the lower mark is awarded to projects in which there is a lack of attention to many of these key features, and the higher mark is awarded to projects in which there is a lack of attention to relatively few of these key features.</li> </ul> </li> <li>• Observations and evaluations of the literature are largely satisfactory but lack detail in one or more aspects.</li> <li>• The source material, field or laboratory work have been measured and recorded accurately, but more care and/or attention to detail are required.</li> <li>• Data collection and presentation fall short of industry/scientific journal standards in one or more areas. The use of technical terminology is sometimes incorrect.</li> <li>• More and/or better quality data and literature could have been reviewed to help support the interpretations made and better use of time could have been made throughout the project.</li> <li>• Few complex interpretations have been made. Interpretations show some weaknesses and/or are not fully supported by the data presented and/or by the relevant literature.</li> <li>• Discussions and Conclusions show evidence of some independent thought, but do not confirm or challenge existing models.</li> <li>• Limited understanding of the work in its wider context has been demonstrated.</li> <li>• Problem-solving skills have been demonstrated, but independent interpretation is limited in scope.</li> <li>• The work contains some irrelevant or inconsistent material, has some issues with structure, shows grammatical inaccuracies and/or does not conform to the assessment guidelines in one or more areas. Some citations are not relevant and more citations are required to support interpretations. Accompanying references show inaccuracies and fall short of the standard for an academic journal.</li> <li>• Some figures used are not relevant, incorporate limited relevant or originally presented content, and/or are of poor quality.</li> </ul> |
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| 40-49 | <ul style="list-style-type: none"> <li>• An <b>Unsatisfactory</b> project will indicate a basic understanding of the methods to be used and how to organise and present the work in the report, but will not have gone beyond this, and there may well be signs of confusion about more complex material.</li> <li>• The work is sound in parts but falls below a satisfactory standard in several areas.</li> <li>• There should be fair work towards the project objectives and the final report must clearly represent a development of some objectives.</li> <li>• A project reporting a basic description of the methodology used to address the problem, and some analysis of the results. However, the project is problematic in terms of key features of the process (e.g. planning, design, rationale, conducting the research, testing, data analysis, critical insight, write-up). <ul style="list-style-type: none"> <li>○ Within this Band, the lower mark is awarded to those projects that are problematic in terms of most of these key features, and the higher mark is awarded to projects that are problematic in terms of relatively few of these key features.</li> </ul> </li> <li>• Only general observations and evaluations of the literature have been made.</li> <li>• The source material, field or laboratory work have been measured and recorded but commonly with insufficient accuracy and/or detail.</li> <li>• Data collection and presentation consistently fall short of industry/scientific journal standards. The use of technical terminology is often incorrect.</li> <li>• Interpretations are very weak and are limited by the amount and/or quality of data collected or literature reviewed. Much better use of time could have been made throughout the project.</li> <li>• Only very general interpretations have been made. Interpretations are very weak and/or not supported by the data presented and/or show very limited knowledge of the relevant literature.</li> <li>• Discussions and Conclusions are commonly inconsistent and do not confirm or challenge existing models.</li> <li>• A poor understanding of the work in its wider context has been demonstrated.</li> <li>• Few problem-solving skills have been demonstrated and independent interpretation is very limited in scope.</li> <li>• The work contains irrelevant and/or inconsistent material and has a confused structure. Grammatical inaccuracies are common and the work does not conform to the assessment guidelines. Inadequate/irrelevant citations have been made. Accompanying references show inaccuracies and fall short of the standard for an academic journal.</li> <li>• Many of the figures and much of the content are not relevant, incorporate negligible originally presented content, and/or are of poor quality.</li> </ul> |
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| 30 - 39 | <ul style="list-style-type: none"> <li>• The work <b>fails</b> to reach an acceptable standard in most areas.</li> <li>• There should be work towards the project objectives, but significant issues are likely to be neglected.</li> <li>• There may be significant errors or misconceptions in the project.</li> <li>• The final report may represent little progress with respect to the interim report.</li> <li>• The project is likely to have substantial shortcomings in crucial features of the design, execution and research process (e.g. planning, design, execution, rationale, conducting the research, data analysis, critical insight, write-up).</li> <li>• Few observations and evaluations of the literature have been made and many of those that have been made are flawed.</li> <li>• Little source material, field or laboratory work has been reviewed and data are inaccurate and/or incorrectly recorded.</li> <li>• Data collection and presentation fall well short of industry/scientific journal standards. The use of technical terminology is very often incorrect.</li> <li>• Interpretations are absent or extremely poor and are severely limited by the amount and/or quality of literature reviewed. Inadequate effort has been put into the project.</li> <li>• Few interpretations have been made and those that have been made show flaws. Interpretations, where present, are very weak and overly reliant upon existing models.</li> <li>• Discussions and Conclusions are inconsistent and do not confirm or challenge existing models. No understanding of the relevant literature has been demonstrated.</li> <li>• No understanding of the work in its wider context has been demonstrated.</li> <li>• Problem-solving skills are very poorly developed and independent interpretation is absent or extremely poor.</li> <li>• The work contains much irrelevant and/or inconsistent material and has a very confused structure. Grammatical inaccuracies are common and the work fails to meet minimum assessment criteria. Inadequate and mostly irrelevant citations have been made. Accompanying references are inaccurate and fall well short of the standard for an academic journal.</li> <li>• Figures used are not relevant, incorporate negligible originally presented content, and/or are of extremely poor quality.</li> </ul> |
| 0 - 29  | <ul style="list-style-type: none"> <li>• The project may contain some correct and relevant material, but most issues are neglected or are covered incorrectly.</li> <li>• There should be some signs of appreciation of the project requirements.</li> <li>• Very little or nothing that is correct and relevant and there is no real appreciation of the project requirements.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |